

8. Getting Results

Now that your data is in R, you can start getting results.

In this section, we will calculate:

- A mean
- A frequency table
- A simple plot

First, import the data again:

```
data <- read.csv("mydata.csv")
```

Calculating a Mean

The **mean** is the average.

Let's calculate the average quiz score:

```
mean(data$quiz_score)
```

```
[1] 81.4
```

The **\$** symbol means:

Use a column from this dataset.

So:

```
data$quiz_score
```

means:

The `quiz_score` column inside the dataset called `data`.

Interpreting the Mean

If R gives you a number such as:

```
[1] 82.4
```

you can write:

The average quiz score is 82.4.

Tip

Always explain results in words. Do not only show the code output.

Calculating Another Mean

You can also calculate the average number of study hours:

```
mean(data$study_hours)
```

```
[1] 6
```

A sentence for this result might be:

Students studied an average of about 6 hours.

Creating a Frequency Table

A **frequency table** counts how many times each category appears.

Let's count the number of students in each major:

```
table(data$major)
```

Biology	Business	Psychology
3	4	3

This tells us how many students are listed for each major.

Interpreting a Frequency Table

If R shows something like:

Biology	Business	Psychology
3	4	3

you can write:

There are 3 Biology students, 4 Business students, and 3 Psychology students in the dataset.

Another Frequency Table

We can count attendance:

```
table(data$attendance)
```

Absent	Present
2	8

This shows how many students were marked as **Present** or **Absent**.

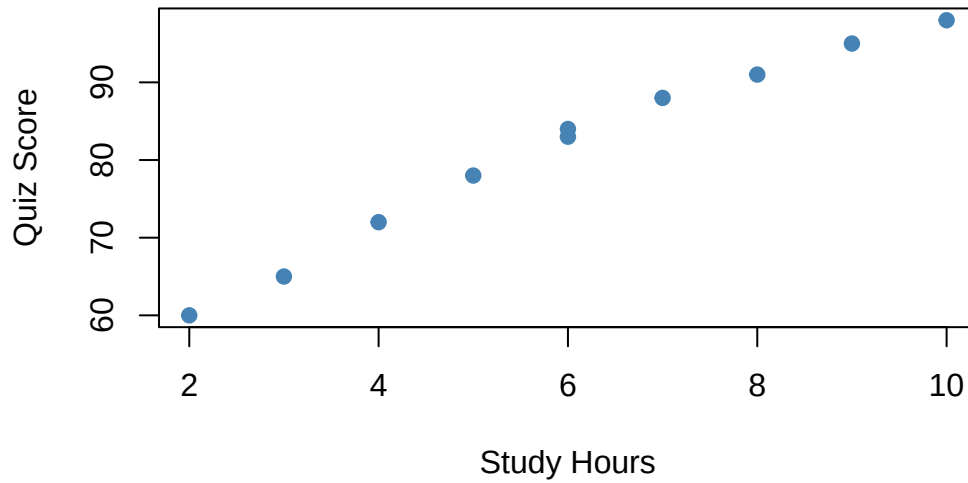
Creating a Simple Plot

A plot helps us see patterns.

Let's make a scatterplot of study hours and quiz scores:

```
plot(  
  data$study_hours,  
  data$quiz_score,  
  main = "Study Hours and Quiz Score",  
  xlab = "Study Hours",  
  ylab = "Quiz Score",  
  pch = 19,  
  col = "steelblue"  
)
```

Study Hours and Quiz Score



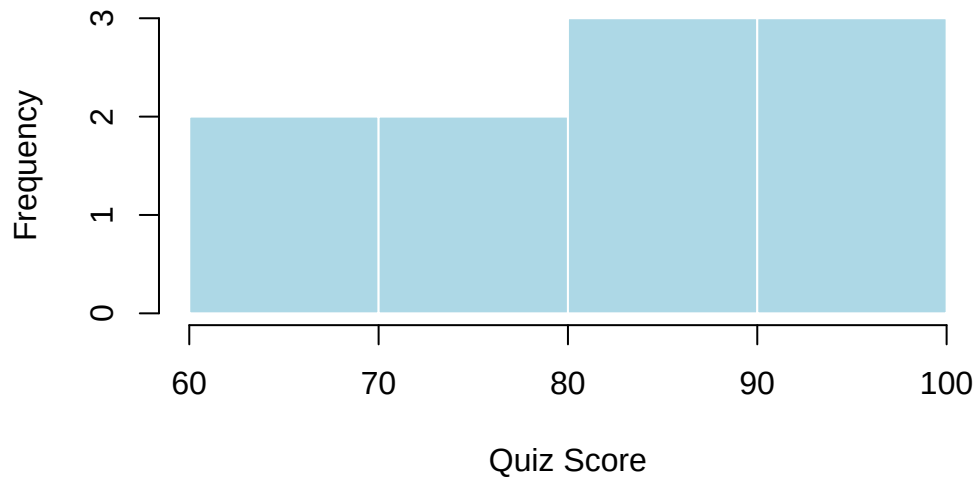
This graph shows whether students who studied more hours tended to have higher quiz scores.

Another Simple Plot

We can also make a histogram of quiz scores:

```
hist(  
  data$quiz_score,  
  main = "Distribution of Quiz Scores",  
  xlab = "Quiz Score",  
  col = "lightblue",  
  border = "white"  
)
```

Distribution of Quiz Scores



A histogram shows how values are spread out.

Try It Yourself

Practice

Use the dataset to answer these questions:

1. What is the average number of study hours?
2. How many students are in each attendance group?
3. Make a plot of sleep hours and quiz score.

Use these commands as a starting point:

```
mean(data$study_hours)
table(data$attendance)

plot(
  data$sleep_hours,
  data$quiz_score,
  main = "Sleep Hours and Quiz Score",
  xlab = "Sleep Hours",
  ylab = "Quiz Score"
)
```

Practice: Write Your Results

Practice

After running your code, write one sentence for each result.

Example:

The average quiz score was 81.4.

Most students in this small dataset were present.

The scatterplot suggests that higher study hours may be related to higher quiz scores.

Writing Results in Sentences

Good results include both the number and an explanation.

Instead of only writing:

```
[1] 6.2
```

write:

The average number of study hours was 6.2 hours.

Instead of only showing a plot, write:

The plot suggests that students who studied more hours tended to have higher quiz scores.

i Reminder

Your goal is not only to run code. Your goal is to understand and explain what the results mean.

Check Your Understanding

Answer these questions:

1. What does `mean()` calculate?
2. What does `table()` create?
3. What does the `$` symbol do?
4. Why should you explain results in words?

Summary

In this section, you learned how to:

- Calculate a mean.
- Create a frequency table.
- Make a simple scatterplot.
- Make a histogram.
- Explain results in plain language.